

# Empirical Analysis for Policymakers

Sciences Po  
Fall 2024

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|--------------------|----------------------------------------------------------------------------------|---------------|-----------------------|
| <b>Instructor:</b> | Emir Tarık Dakın                                                                 | <b>Time:</b>  | Tuesdays, 16:40-18:40 |
| <b>Email:</b>      | <a href="mailto:emirtarik.dakin@sciencespo.fr">emirtarik.dakin@sciencespo.fr</a> | <b>Place:</b> | I202                  |

## Academic integrity principles

Sciences Po's educational project encourages students to develop **independent and critical thinking, intellectual honesty and respect for copyright and image rights** in the use of sources.

To help you meet these requirements and consolidate your learning, we invite you to refer to the following documents and resources:

[Academic rules and regulation \(article 12\)](#);  
[Academic integrity charter](#);  
[Mandatory academic integrity online training](#) on Moodle;  
[Methodological resources](#) on the student website.

We remind you that any fraud or proven attempted fraud will be punished in accordance with the [disciplinary procedures](#) in force at Sciences Po.

**Office Hours:** After class, or by appointment, or post your questions in the forum provided on Moodle.

## Textbook:

- Joshua Angrist and Jorn-Steffen Pischke, *Mastering 'Metrics: The Path from Cause to Effect*. Princeton University Press, 2014. ISBN-13: 978-0691152844

Along with this book, you will be assigned mandatory readings of academic articles. The readings will be available for download in PDF format from Moodle. The Angrist/Pischke book follows the readings. For many of the major sections in the course, we will read an academic article that illustrates the techniques used in class. The Angrist/Pischke book will explain some of the readings.

**Course Description:** This course introduces inferential statistical methods and econometric/regression analysis tools essential for understanding empirical public policy and planning research and for conducting basic applied policy analysis. Statistical concepts are demonstrated using data and examples mainly from public policy and planning. You will learn to utilize quantitative data analysis tools to examine empirical evidence of real-world events.

The study of economics, planning, and other social sciences involves both theoretical and empirical approaches. Theory involves studying models of the real world based on

simplifying assumptions, predicting outcomes based on these assumptions. Empirical social science seeks to validate these theoretical relationships by analyzing data on events and relationships using mathematical and statistical techniques, thereby testing the validity of theoretical models. Inferential statistics and econometrics encompass the tools and techniques for performing this empirical analysis.

By the end of this course, you should be able to formulate an empirically testable research question, conduct basic empirical research on this question using statistical software, and read and understand empirical literature in relevant field journals.

**Objectives:** This course offers an opportunity to conduct original empirical research using R and enhance your skills in written and graphical communication. The primary aim is to help you gain significant proficiency in inferential statistics, a crucial quantitative method in professional planning and policy practice.

Moreover, performing high-quality empirical research necessitates an understanding of professional ethics. Depending on your chosen project for this course, you will have the chance to use empirical methods to investigate important policy and planning issues, including growth and development, social justice, sustainability, and environmental concerns.

**Attendance:** Attendance is required for this course. If you absolutely must miss a class, please let me know. Bringing your laptops to class are highly recommended.

**Prerequisites:** A course in Statistics and/or Probability Theory is required. You are also expected to know some simple calculus.

**Statistical Software:** We will be using R extensively throughout this course. R is a software used in academics to build statistical models and derive causal inferences from data. You will possibly need to use R for your thesis and definitely need it for your future academic life and/or to get a job.

To get started with R, please have a look at the [Principles of Econometrics with R \(Colonescu, 2016\)](#) as we will be following this guide, especially in the beginning of the course.

**Use of Generative AI:** You are free to use generative AI for coding and understanding concepts. However, you are not allowed to use AI for generating text to be used in your assignments. Turned in assignments with clear AI wording and explanations will receive a grade of zero and are subject to disciplinary action for plagiarism. Please try to use your own words as much as possible or provide citations to relevant academic articles where necessary.

### Grading Policy:

- **Quizzes (30%)**

There will be four short quizzes that will be announced beforehand. These will be comprised of short questions or MCQs about the content we have covered up until the quiz of that session. Missed quizzes will receive a grade of zero unless a doctor's notice is provided, in which case there will be a chance to retake the quiz in the following week.

- **Homework** - *in groups of two* (20%)

You will be assigned three homework assignments throughout the semester, in which you have to do R exercises related to the concepts we cover in class. You are expected to turn in a page or two of text of your work, plus your R script which is separate from the text.

- **Empirical project** - *in groups of three* (50%)

You will be provided with a selection of data sets and fairly narrow research questions and you will be expected to review the relevant literature, estimate models to answer the particular question, and write up the results as if this were an academic paper. The assignments will be distributed on October 15th.

You will present your work in short sessions (about 20 minutes) which will be followed by questions. At the end of the semester, you will turn in a short replica of an academic paper, which will include your story, data/methodology and your findings. Grades will be based on the quality of the presentation, your oral defense and ability to answer questions, quality of the writing of the paper, the justification for the model you estimate, and the accuracy with which you interpret your statistical models.

**Important Dates:**

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|--------------------------|-----------------|
| Project assignment ..... | 15/10           |
| Presentations .....      | 19/11 and 26/11 |
| Paper deadline .....     | 10/12           |

## Outline

- I. **Overview and Introduction to R**
- II. **Introduction to R (cont'd)**
- III. **R Workshop**
- IV. **Basic probability theory and regressions: OLS**
  - Textbook Chapter 2 - Regressions
  - Textbook Chapter 2 Appendix - Regression Theory
- V. **OLS (cont'd)**
  - Textbook Chapter 2.1 - A Tale of Two Colleges
- VI. **Instrumental Variables**
  - Textbook Chapter 3.2 - Abuse Busters
  - Acemoglu, D., Johnson, S., and Robinson, J. A. (2001). The Colonial Origins of Comparative Development: An Empirical Investigation. *American Economic Review*, 91(5), 1369–1401. <http://www.jstor.org/stable/2677930>
- VII. **Instrumental Variables (cont'd)**
  - Angrist, J.D. “Instrumental variables methods in experimental criminological research: what, why and how.” *J Exp Criminol* 2, 23–44 (2006).  
<https://doi.org/10.1007/s11292-005-5126-x>
- VIII. **Difference-in-Differences**
  - Textbook Chapter 5.2 - Drink, Drank, ...
  - Norberg KE, Bierut LJ, Gruzka RA. “Long-term effects of minimum drinking age laws on past-year alcohol and drug use disorders”. *Alcohol Clin Exp Res*. 2009 Dec; 33(12):2180-90. <https://doi.org/10.1111/j.1530-0277.2009.01056.x>
- IX. **Regression Discontinuity**
  - Textbook Chapter 4.1 - Birthdays and Funerals
  - Carpenter, Christopher, and Carlos Dobkin. 2011. “The Minimum Legal Drinking Age and Public Health.” *Journal of Economic Perspectives*, 25 (2): 133–56. <https://doi.org/10.1257/jep.25.2.133>
- X. **Review**
- XI. **Presentations**
- XII. **Presentations**